Enrollment No:

End-Term Examination
(CBCS)(SUBJECTIVE TYPE)(OffLine)
Course Name: B.Tech. (IT, CSE, CSE-AI), Semester:<3>
(Nov-Dec, 2022)

Subject Code: BCS 203 **Subject: Discrete Structure**
Time :3 Hours **Maximum Marks :60**
Note:Q. 1 is compulsory. Attempt one question each from the Units I, II, III & IV.

Q1 **(2.5*8=20)**

(a) Write the following propositions in symbolic form
(i) Every computer is a 16 bit-machine
(ii) Every integer is having a greater positive integer

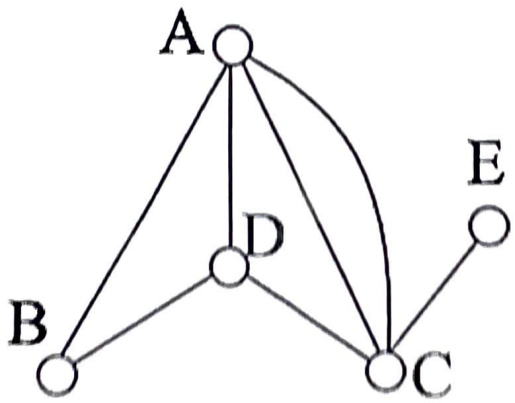
(b) Draw the Hasse diagram of f (P(A),⊆) where P(A) is the power set of A and A={a,b,c,d} .

(c) Let N be the set of all natural numbers. For each of the following determine whether * is an associative operation
(i) a*b= Max(a,b)
(ii) a*b=Min(a,b+2)

(d) How many edges are there in a graph 10 vertices each of degree six? Give an application of Hamiltonian Path

(e) Find the no of positive Integers ≤ 2076 and is divisible by 3,5 and 7.

(f) Examine and justify your answer whether the following graph has a . Euler path
b. Euler Circuit



(g) Find the composition (*gof*)(x) and (fog)(x) where $f(x) = 2x$ and $g(x) = x^2+3$

(h) Determine the truth value of each of these statements the domain consisting of real numbers
a. $\exists(x)\ (x^2=2)$
b. $\exists(x)\ (x^2=-1)$

UNIT-I

Q2 **(2.5,2.5,5)**
a. Using quantifiers to express the statement that " there does not exist a student who has taken a flight every airline in the world"
b. Translate the following into English statement
$V(x)\ V(y)\ ((x>0)\wedge(y<0) \rightarrow (xy<0))$
c. Prove by theory of inference that the following argument is correct
If today is Tuesday, I have a test of Mathematics or Economics. If my Economics professor is sick,
I will not have a test of Economics. Today is Tuesday and my Economics professor is sick, therefore, I have a test of Mathematics.

PTO

Q3 Prove that if n is an integer and 3n+2 is even, then n is even by using (10)

a. A Proof by Contraposition
b. A proof by contradiction

UNIT-II

Q4 (a) Prove that the relation Divides on Integers is a POSET. Construct the hasse diagram of (D_{45} , Divides), where D_{45} stands for all the factors of 45. Find the least member and highest member of the above POSET (5+2+3)

(b) A C++ identifier contains 37 alphanumeric characters. Show that at least two characters are same

(c) Prove that, $\neg(p \vee (\neg p \wedge q))$ is logically equivalent to $\neg p \wedge \neg q$ without using truth table

Q5 a. Show that the relation "x R y $\Leftrightarrow$ x – y is divisible by 3" where x, y in the set of integer , is an equivalence relation. (10)

b. Write the recurrence relation that represents a merge sort. Solve this recurrence relation to find the complexity of Merge Sort

UNIT-III

Q6 a. Prove that intersection of two subgroups of a group (G,*) is a subgroup of a group (G,*) (5,5)

b. Show that if every element of the group G is its own inverse, then G is abelian

Q7 a. Show that the set {5, 15, 25, 35} is a group under multiplication modulo 40. (3,3,4)

b. Let x belong to a group. If $x^2 \neq e$ while $x^6 = e$, prove that $x^4 \neq e$ and $x^5 \neq e$. What is the order of x? (where e is the identity element)

c. Define Ring, Field and Cyclic Group

UNIT-IV

Q8 a. State and prove Euler's formula for a plane connected graph (4,2,4)

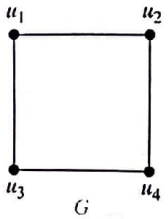
b. How can you determine if a given graph contains an Euler path?

c. Prove that C_6 is bipartite

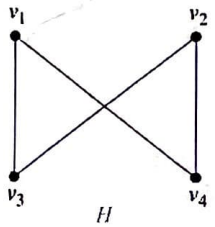
Q9 a. Proof four color theorem for planer graph (4,2,4)

b. What is the chromatic number of a tree of 20 Nodes?

c. Show that the graphs $G = (V, E)$ and $H = (W, F)$ are isomorphic where



G



H

570